

Physiologic Mechanisms in Aortic Insufficiency

I. The Effect of Changing Heart Rate on Flow Dynamics

II. Determinants of Austin Flint Murmur

SHLOMO LANIADO, M.D., EDWARD L. YELLIN, Ph.D., CHAIM YORAN, M.D.,

JOEL STROM, M.D., MASATSUGU HORI, M.D., SHLOMO GABBAY, M.D.,

REUVEN TERDIMAN, M.Sc., AND ROBERT W.M. FRATER, M.B., Ch.B.

SUMMARY We studied the dynamic changes in mitral flow patterns and in mitral valve motion before and after producing acute, reversible aortic insufficiency (AI) in nine open-chest dogs. Phasic mitral flow, the mitral valve echocardiogram, the intracardiac phonocardiogram and other hemodynamic variables were measured. During moderate AI (mean regurgitant fraction $52 \pm 5\%$) ($\pm$ SD), the antegrade filling volume decreased from 31 ± 7 to 24 ± 6 ml ($p < 0.01$), but the peak protodiastolic mitral flow rate increased from 139 ± 37 to 157 ± 42 ml/sec ($p < 0.01$), reflecting the shift of a larger fraction of total mitral filling volume to early diastole. In six dogs, atrial pacing was used to examine the hemodynamic effects of tachycardia. Increasing the heart rate from 90 to 120 beats/min increased cardiac output from 2.64 ± 0.56 to 3.3 ± 0.831 l/min ($p < 0.05$) and decreased left atrial pressure from 24 ± 8 to 17 ± 7 mm Hg ($p < 0.05$). Increasing heart rate to 150 beats/min compromised mitral filling, reduced cardiac output and increased left atrial pressure. Moderate tachycardia improves cardiac performance in AI by reducing regurgitant volume, without significantly reducing transmitral filling volume. The mitral valve echocardiogram showed only a small decrease in cusp opening amplitude during AI. A low-pitched left ventricular inflow tract murmur was recorded in protodiastole and corresponded in time to the rapidly increasing mitral flow. We conclude that the major determinant of the turbulence responsible for the creation of the Austin Flint murmur is the antegrade mitral flow stream and its mixing with the retrograde aortic flow.

THE PATHOPHYSIOLOGY, clinical recognition and management of aortic insufficiency (AI) have been studied extensively in recent years. Clinical expressions of chronic and acute aortic regurgitation have been well defined,¹ and data regarding the natural history of the disease^{2,3} and the proper selection of patients for medical vs surgical treatment is rapidly accumulating.⁴⁻⁷ However, little is known about the dynamics of atrioventricular diastolic flow transport and, hence, about the significance of alterations in diastolic left ventricular (LV) filling in AI. To compensate for the immediate decrease in effective flow, AI may rapidly activate other physiologic mechanisms in addition to the Starling mechanism. Tachycardia is of particular benefit to patients with chronic or acute AI,⁸ although the mechanism by which increased heart rate (HR) specifically favors the hemodynamics of AI is not clear.⁹⁻¹²

We undertook the present study to investigate the dynamic changes that occur on the left side of the heart with the onset of acute AI. Using our established technique of simultaneous phasic mitral flow and echocardiographic measurements, we placed particular emphasis on LV filling and mitral valve motion during the

different phases of diastole. We concentrated on two aspects of mitral flow: the mechanism by which moderate increases in HR affect mitral and aortic flow to improve LV pump function, and the role of mitral flow in the production of the Austin Flint murmur.

Methods

Animal Preparation and Instrumentation

Nine healthy mongrel dogs that weighed 25–30 kg were anesthetized with i.v. pentobarbital, 30 mg/kg, and placed on artificial respiration. The chest was opened by midsternotomy and left thoracotomy, the pericardium was incised, and the heart was suspended in a pericardial cradle. Cardiac instrumentation was carried out as described previously.¹³⁻¹⁵ Briefly, transmitral flow was recorded with an electromagnetic flow probe sutured above the mitral annulus during cardiopulmonary bypass. A second flow probe was positioned around the proximal ascending aorta.

High-fidelity pressure readings were obtained by inserting catheter-tip micromanometers (Millar) into the left ventricle and left atrium. The aortic pressure was measured with a fluid-filled catheter and a Statham gauge. All three pressures were calibrated for equal sensitivity and common zero. The rate of rise of LV pressure (LV dP/dt) and the LV intracardiac phonocardiogram were derived from the high-fidelity LV pressure signal.¹³ When the phonocardiogram was recorded, the transducer was located at the LV inflow tract. M-mode or two-dimensional echocardiograms of the mitral valve or LV structures were obtained by lightly placing the transducer on the right ventricular surface adjacent to the interventricular septum.¹⁴

Reversible AI was induced with a collapsible basket carried at the tip of a catheter. The basket was intro-

From the Department of Surgery Cardiovascular Research Laboratory and the Departments of Physiology and Medicine, Albert Einstein College of Medicine, Bronx, New York, and the Department of Cardiology, Tel-Aviv Medical Center (Ichilov Hospital) and Sackler School of Medicine, Tel-Aviv, Israel.

Supported by PHS grants HL-19391 and HL-24638.

Presented in part at the American College of Cardiology Annual Meeting, March 1981.

Address for correspondence: Edward L. Yellin, Ph.D., Albert Einstein College of Medicine, Yeshiva University, 1300 Morris Park Avenue, Bronx, New York 10461.

Received August 3, 1981; revision accepted November 6, 1981.

Circulation 66, No. 1, 1982.

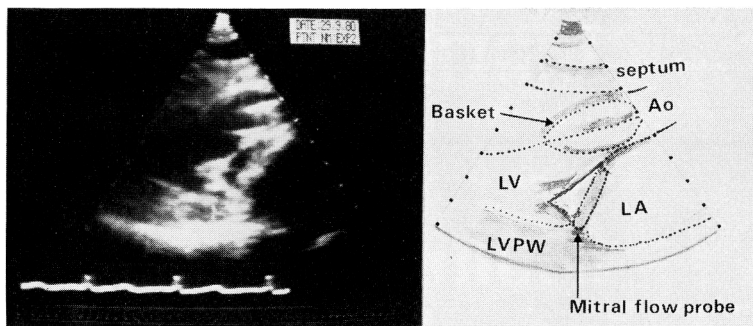


FIGURE 1. A frame taken from a two-dimensional echocardiogram showing a longitudinal view of the mitral valve with the basket in an open position in the aortic valve. The mitral flow probe is seen in profile (arrow, right panel). Ao = aorta; LV = left ventricle; LA = left atrium; LVPW = left ventricular posterior wall.

duced through the aortic valve through the LV apex or through the carotid artery (fig. 1), and was opened and closed by remote control.^{16, 17} To investigate the effects of increased HR on the hemodynamics of AI, in six dogs the sinoatrial node was crushed and the left atrium was paced at rates of 90, 120 and 150 beats/min. Atrial pacing was also used to achieve a constant HR when regular sinus rhythm was absent.

Measurements and Calculations

The hemodynamic data and ECG were recorded on an oscillographic recorder (DR-12, Electronics for Medicine) at 100 mm/sec. M-mode echocardiograms were recorded using a separate echocardiograph (Picker, model 80 C1). To synchronize different events accurately, the aortic flow curves were registered simultaneously by both recorders. Two-dimensional echocardiograms were recorded on a video tape recorder (Sanyo 7100) and events were synchronized by the ECG and audio input.

Aortic stroke volume (SV), aortic regurgitant vol-

ume (RV) and LV filling volume (FV) (fig. 2) were calculated by digitizing the respective aortic and mitral flow curves with a sonic digitizer (Science Accessories GP-3) coupled to a digital computer (PDP 11/34). The accuracy of the method has been verified.¹⁵ Regurgitant fraction (RF) was calculated as RV/SV ; cardiac output as $(SV - RV) \times HR = FV \times HR$; total ejected volume (TEV) as $SV \times HR$; and total regurgitant volume (TRV) as $RV \times HR$.

The zero level for mitral flow was determined during systole and for aortic flow during diastole before the induction of AI. Occasionally, the opening of the basket apparently shifted the aortic flow curve and a new baseline was established as follows. For each test sequence to be analyzed, the average FV of three control (before AI) cardiac beats was divided by the average of a similar number of aortic SVs. The ratio was defined as a correction factor to the aortic flow calibration. The aortic flow baseline during regurgitation was then determined by setting $FV = SV - RV$. FV was chosen as the standard because the mitral flow probe is

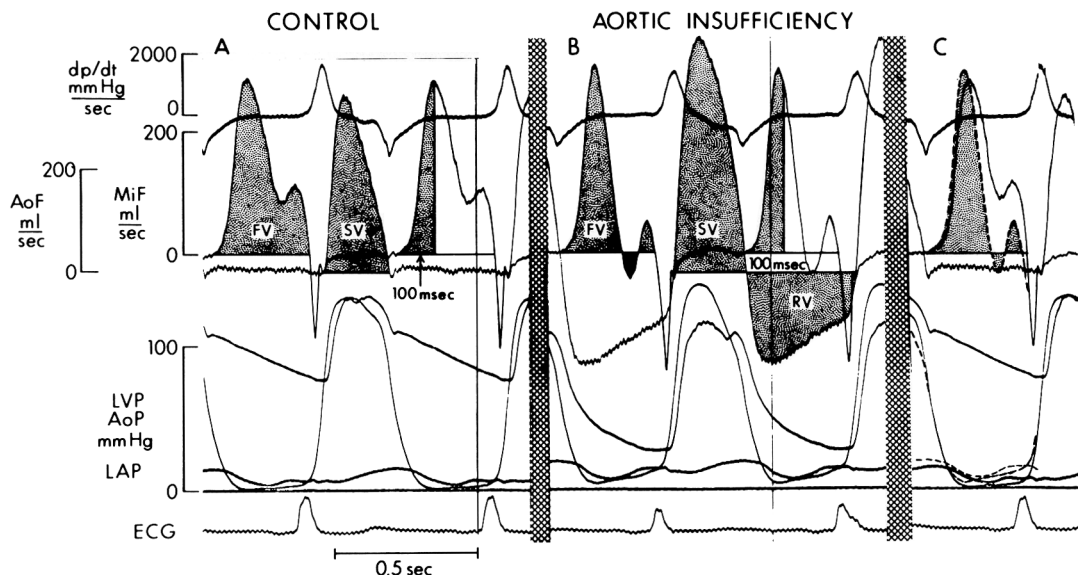


FIGURE 2. Original oscillographic record of hemodynamic data before aortic insufficiency (A) and during steady-state aortic insufficiency (B). (C) Records before and during aortic insufficiency are superposed to illustrate changes in mitral flow (MiF) configuration, left ventricular pressure (LVP) and left atrial diastolic pressure (LAP). During aortic insufficiency, total mitral filling volume (FV) decreased, but FV during the first 100 msec of diastole and maximal MiF velocity both increased. A small amount of mid-diastolic mitral regurgitation is evident during aortic insufficiency (solid black area under mitral flow in panel B). AoF = aortic flow; SV = total aortic stroke volume; RV = regurgitant volume; AoP = aortic pressure. ECG lead II is shown.

TABLE 1. Control and Steady-state Hemodynamic Results After Production of Aortic Insufficiency

Dog	HR (beats/min)	CO (l/min)	RF (%)	AoP (mm Hg)				Peak LVP (mm Hg)	
				Systolic		Diastolic		C	AI
				C	AI	C	AI		
1 (n = 9)	100 ± 6	2.33 ± 0.36	57 ± 3	87 ± 7	100 ± 4	44 ± 6	26 ± 9	87 ± 7	100 ± 36
2 (n = 6)	98 ± 8	2.76 ± 0.34	50 ± 4	109 ± 17	115 ± 20	59 ± 5	41 ± 6	112 ± 10	120 ± 22
3 (n = 10)	108 ± 6	3.05 ± 0.54	59 ± 7	87 ± 16	88 ± 14	44 ± 12	26 ± 4	89 ± 10	96 ± 6
4 (n = 7)	104 ± 4	2.86 ± 0.24	56 ± 8	85 ± 10	94 ± 16	44 ± 8	23 ± 2	86 ± 5	99 ± 6
5 (n = 7)	92 ± 8	2.67 ± 0.61	48 ± 2	116 ± 9	130 ± 12	60 ± 20	40 ± 6	115 ± 1	130 ± 24
6 (n = 5)	95 ± 7	3.72 ± 0.98	50 ± 5	122 ± 5	132 ± 6	52 ± 12	31 ± 5	122 ± 4	132 ± 6
7 (n = 11)	93 ± 5	4.09 ± 1.03	52 ± 3	130 ± 16	130 ± 13	70 ± 9	35 ± 9	140 ± 9	140 ± 3
8 (n = 7)	98 ± 7	3.67 ± 0.93	53 ± 4	103 ± 10	112 ± 11	68 ± 6	36 ± 3	113 ± 8	128 ± 5
9 (n = 9)	88 ± 5	3.21 ± 0.90	42 ± 3	100 ± 7	115 ± 4	56 ± 7	28 ± 9	101 ± 5	118 ± 13
Average (n = 71)	97 ± 6	3.15 ± 0.57	52 ± 5	104 ± 16	113 ± 16	55 ± 10	32* ± 6	107 ± 18	118 ± 16

All values are mean $\pm$ SD. The diastolic period was measured from the duration of mitral flow; the systolic period is the balance of the cycle.

* $p < 0.01$ vs control (by analysis of variance).

Abbreviations: HR = heart rate; CO = cardiac output; RF = regurgitant fraction; AoP = aortic pressure; LVP = left ventricular pressure; LAP = left atrial pressure measured at onset of filling; FV = filling volume; FV100 = filling volume at 100 msec per beat; MiF = mitral flow; AI = aortic insufficiency; C = control; n = number of runs.

in direct contact with the blood, making it inherently more accurate than the cuff-type aortic flow probe. With the newly adjusted baseline, FV equalled effective SV during the steady state.

The diastolic period was defined as the time between the onset and end of mitral flow. The remainder of the cycle was defined as the systolic period. Time was measured with the resolution of 5 msec (0.5 mm at a paper speed of 100 mm/sec). To compare changes in mitral flow patterns before and during AI, the FV during the first 100 msec of diastole was calculated.

Statistical significance was determined by analysis of variance and t test for grouped data or by paired t test for individual changes during pacing. Differences were considered significant at the $p < 0.05$ level. Data are mean $\pm$ SD.

Results

The amount of induced aortic regurgitation varied from physiologically insignificant to 70%. Because a regurgitant fraction less than 35% produced negligible immediate physiologic alterations in atrioventricular flow transport, we analyzed only experimental sequences with regurgitant fractions greater than 35%.

Hemodynamic Changes

The steady-state hemodynamic changes (achieved within 10 beats after opening the basket) are summarized in table 1 and illustrated in figures 2–5. The HRs shown in table 1 are all normal sinus rhythm and are either the baseline value for the dog or the result of

slowed sinoatrial node activity due to mechanical crushing. The control HR, cardiac output, and pressures were all in the physiologic range. AI produced regurgitant fractions averaging 52%. The ratio of AI to control FV was 77%. The dogs tolerated this level of lesion severity for the duration of each run, which lasted at most approximately 1 minute for the changing HR procedure. Typical steady-state hemodynamic changes after the creation of AI are shown in figure 2.

After severe AI was induced, LV, aortic and left atrial pressures reached similar levels during diastole (fig. 3). LV diastolic and left atrial filling pressures also increased significantly.

Figure 4 is an oscillographic record from an experiment in which the control left atrial pressure was greater than that in figures 2 and 3. During AI, antegrade LV filling was completed before the first half of diastole. During the rest of diastole, the left ventricle was filled solely from retrograde aortic flow.

Table 1 is a summary of the hemodynamic results in nine dogs. In steady-state AI, the left atrial filling pressure was greater than control (92%, $p < 0.01$), resulting in a greater left atrial–LV pressure gradient in early diastole. The increased gradient caused mitral flow to accelerate more rapidly to a peak that was higher than control (13%, $p < 0.01$). Thereafter, a sharp decline in the atrioventricular pressure gradient decelerated mitral flow to zero or sometimes a negative level (figs. 2 and 3). The increase in early-phase peak mitral flow during the steady state partially compensated for the immediate decrease in total FV observed in

TABLE 1. (Continued)

LAP (mm Hg)		Period (msec)				FV (ml)		FV100 (ml)		Peak MiF (ml/sec)	
C	AI	Systolic		Diastolic		C	AI	C	AI	C	AI
9	16	212	272	328	328	23	20	8	9	152	170
±4	±6	±13	±27	±13	±27	±1	±1	±5	±3	±32	±38
7	11	280	313	332	280	28	27	11	13	183	199
±1	±4	±8	±13	±8	±8	±3	±5	±2	±1	±73	±53
10	23	280	285	275	271	28	23	15	13	192	231
±2	±7	±36	±14	±36	±14	±1	±6	±2	±6	±56	±28
11	23	287	276	290	301	27	22	14	14	167	180
±1	±2	±9	±17	±9	±17	±2	±2	±4	±2	±24	±15
11	19	298	304	354	349	28	18	9	10	103	121
±1	±3	±13	±7	±13	±7	±2	±2	±3	±1	±10	±14
9	18	286	274	346	357	32	23	14	17	89	101
±1	±1	±7	±14	±7	±7	±3	±3	±1	±4	±21	±9
12	24	295	301	386	345	44	35	14	13	121	134
±3	±3	±18	±7	±18	±24	±5	±3	±6	±2	±21	±6
16	30	285	308	327	304	37	18	15	17	131	142
±1	±1	±7	±23	±7	±23	±2	±3	±6	±5	±29	±28
7	15	309	345	373	336	37	30	12	11	109	129
±2	±1	±6	±22	±6	±22	±3	±6	±9	±2	±25	±22
10	20*	288	298	331	319	31	24*	12	13	139	157*
±3	±6	±11	±24	±31	±31	±7	±6	±3	±3	±37	±42

the beats immediately after the creation of AI (23%, $p < 0.01$). Figures 2–4 and table 1 demonstrate the temporal shift in diastolic atrioventricular flow transport after AI. A larger fraction of the total transmitral FV was shifted to the early phase of diastole. The changes in the temporal distribution of mitral flow and FV, and the relative contribution of the rapid filling phase to the total FV, have been quantified by arbitrarily dividing the total diastolic filling period into two parts — the first 100 msec and the rest of diastole (fig. 2, table 1). Before AI, 39% of the mitral FV entered the left ventri-

cle during the first 100 msec, but during AI, the ratio was reversed to 54% ($p < 0.01$).

Since a considerable amount of the total mitral filling is shifted to an earlier phase of diastole in AI, one can presume that slow HRs with long diastolic periods will selectively favor the filling of the left ventricle from the aortic (backward) source. In contrast, increasing the HR and shortening diastole should decrease retrograde flow relatively more than antegrade flow, thereby decreasing the regurgitant fraction and increasing the effective cardiac output. This concept

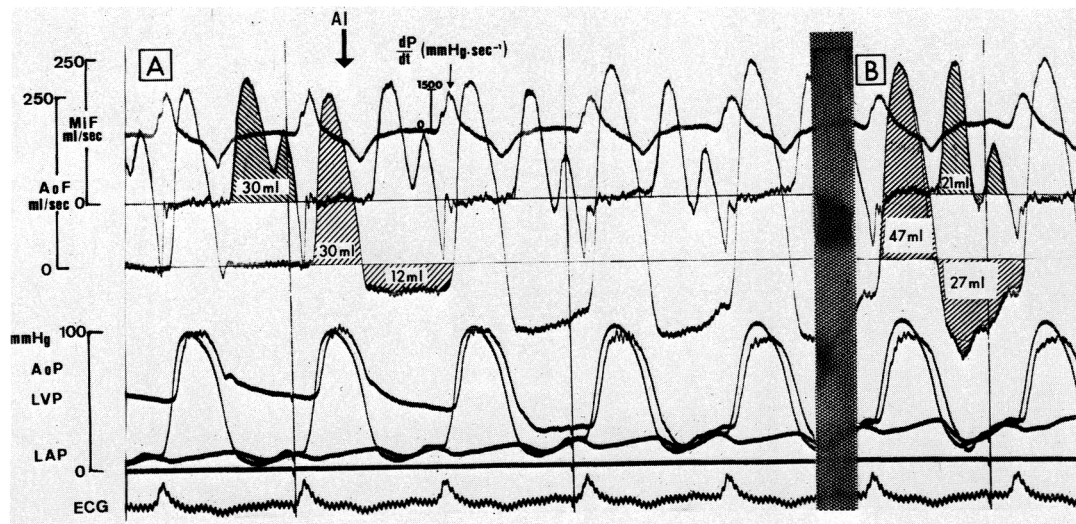


FIGURE 3. Oscillographic record illustrating the transient (A) and steady-state (B) response to severe acute aortic insufficiency (AI) (regurgitant fraction 67%). With the creation of aortic insufficiency, there was an immediate increase in left ventricular diastolic pressure (LVP) and left atrial pressure (LAP). Early in diastole, LVP, aortic pressure (AoP) and LAP were at a similar level; phasic differences accounted for the variations in flow configurations. Time lines indicate 1 second. AoF = aortic flow; MiF = mitral flow.

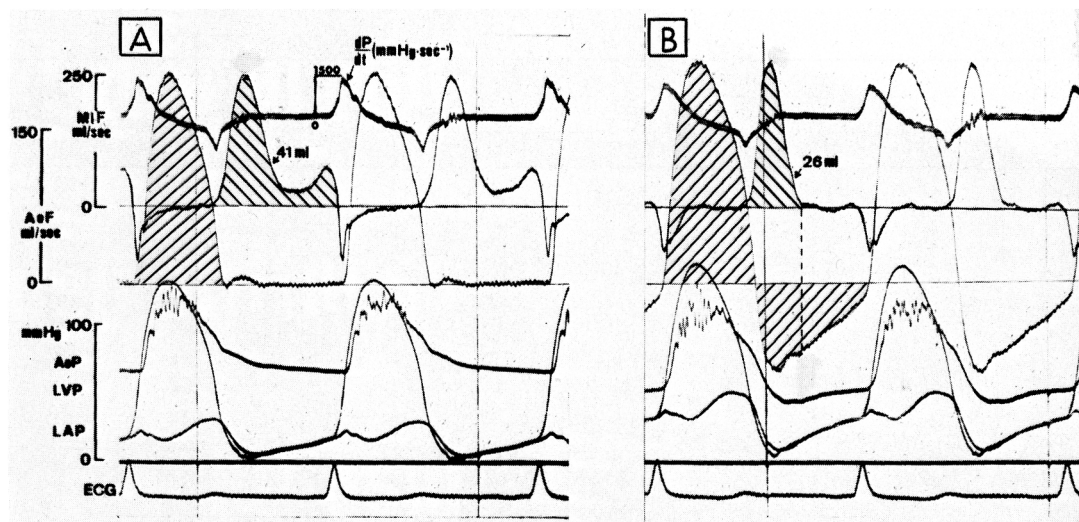


FIGURE 4. (A) Oscillographic record illustrating severe aortic insufficiency in a dog with control left ventricular filling pressure (LVP) greater than that shown in figures 2 and 3. During aortic insufficiency, the total diastolic LV filling was confined to the early phase of diastole (broken line, panel B). The rest of diastole did not contribute to transmitral ventricular filling. Time lines indicate 1 second. LAP = left atrial pressure; MiF = mitral flow; AoF = aortic flow.

was validated by changing the HR in a stepwise manner by atrial pacing during AI in six dogs.

Figure 5 shows the hemodynamic changes in flow patterns and the calculated values of SV, RV and FV in an experiment with HRs of 87, 120, and 150 beats/

min. During AI, as HR increased from 87 to 120 beats/min, effective cardiac output increased significantly (2.7 to 3.4 l/min) with a decrease in LV end-diastolic and left atrial pressures; stroke volume decreased (23%), but the total forward minute output increased

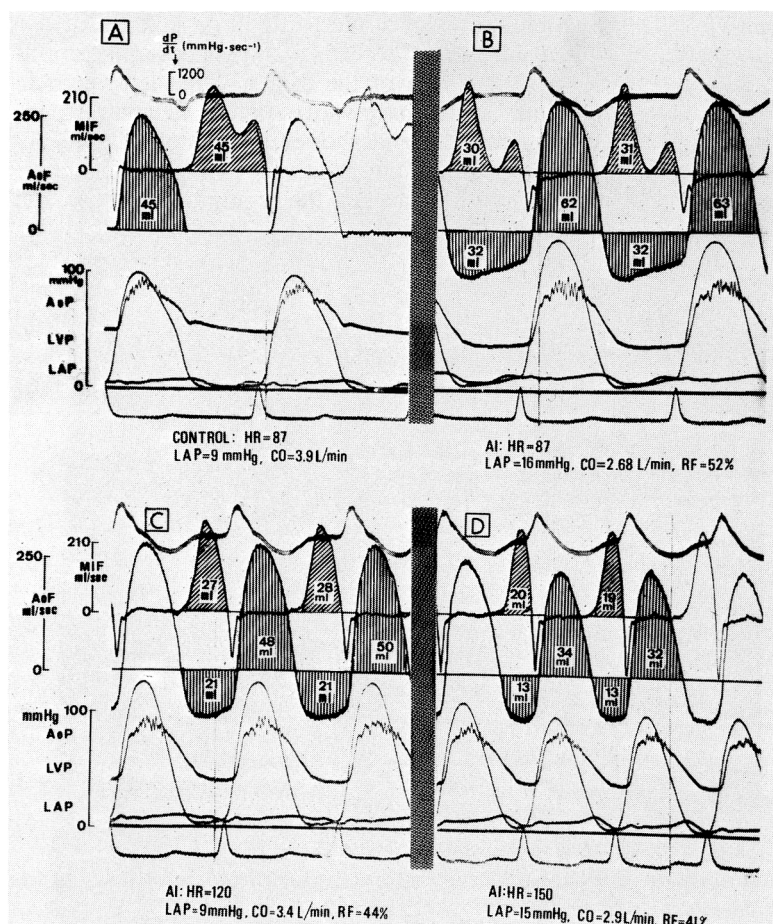


FIGURE 5. Oscillographic records illustrating the effect of increasing heart rate (HR) after creation of aortic insufficiency (AI). The optimal cardiac output (CO) and left atrial pressure (LAP) during AI were achieved at 120 beats/min. MiF = mitral flow; AoF = aortic flow; LVP = left ventricular pressure; AoP = aortic pressure; RF = regurgitant fraction.

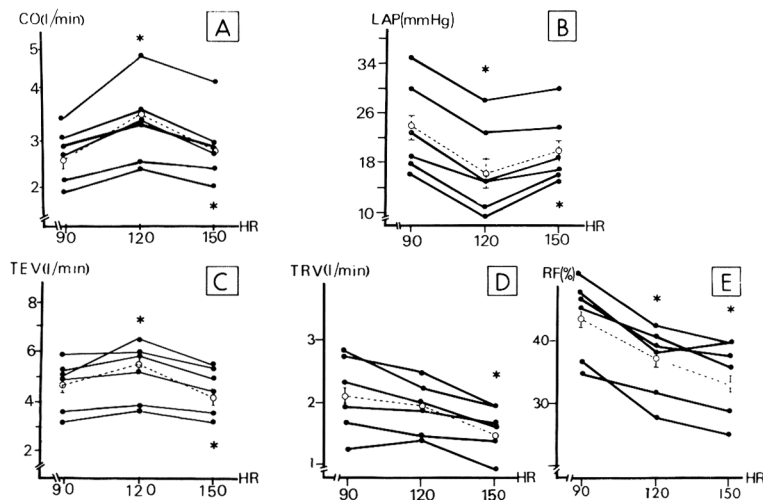


FIGURE 6. Hemodynamic changes in six dogs after acute aortic insufficiency as heart rate (HR) was increased from 90 to 120 and 150 beats/min. Broken lines connecting open circles indicate mean $\pm$ SD. Asterisk indicates $p < 0.05$ by paired t test (vs HR of 90 beats/min in upper part of each panel; vs HR of 120 beats/min in lower part). CO, total ejected volume (TEV), total regurgitant volume (TRV) are minute values. RF = regurgitant fraction.

slightly (5.4 to 5.8 l/min) and regurgitant fraction decreased (52% to 44%). Increasing HR to 150 beats/min decreased the length of diastole to a degree where both mitral and regurgitant flow were compromised; total filling and cardiac output decreased and LV end-diastolic pressure increased. Figure 6 shows the effects of increased rate on the hemodynamics in the six dogs subjected to atrial pacing. Increasing the HR from 90 to 120 beats/min increased cardiac output (2.64 ± 0.56 to 3.30 ± 0.83 l/min, $p < 0.05$), reduced left atrial pressure (24 ± 8 to 17 ± 7 mm Hg, $p < 0.05$) and increased ejected volume per minute (4.76 ± 1.04 to 5.18 ± 1.15 l/min, $p < 0.05$), but did not significantly change the RV per minute (2.12 ± 0.66 to 1.90 ± 0.48 l/min) and decreased regurgitant fraction ($44 \pm 7\%$ to $37 \pm 6\%$, $p < 0.05$). These results occurred because when HR was increased from 90 to 120 beats/min, the FV decreased by only 8% (29 ± 6 to 27 ± 7 ml, $p < 0.05$), while RV decreased by 33% (24 ± 7 to 16 ± 4 , $p < 0.001$). Although total RV and regurgitant fraction both decreased at HRs of 150 beats/min, their beneficial effects were offset by the decrease in cardiac output and increase in left atrial filling pressure. In these experiments, optimal hemodynamics in AI were achieved at HRs of about 120 beats/min.

Echocardiographic Observations

Two-dimensional echocardiography was used to verify that the basket did not interfere with full opening motion of the mitral valve (figs. 1 and 7). Occasional fine flutter of the septum and mitral valve was observed, and LV chamber dimensions increased after acute AI was produced.

When both cusps could be sufficiently delineated by M-mode echocardiography during AI, distance between the tips of the opened cusps was reduced. The amplitude of opening was reduced 10–20%, and at mid-diastole the cusps came closer to each other (fig. 7). However, the reduction in cusp amplitude was not seen in all the experiments.

Figure 8 shows a tendency for the cusps to move toward premature but incomplete closure of the mitral valve. This was more common in experiments with severe regurgitation, signs of failure, low cardiac output, and high LV end-diastolic pressure. In figure 8, LV depression before induction of AI was due to myocardial ischemia.

Phonocardiographic Observations

Figures 9 and 10 show phonocardiograms derived from a high-fidelity transducer located at the inflow

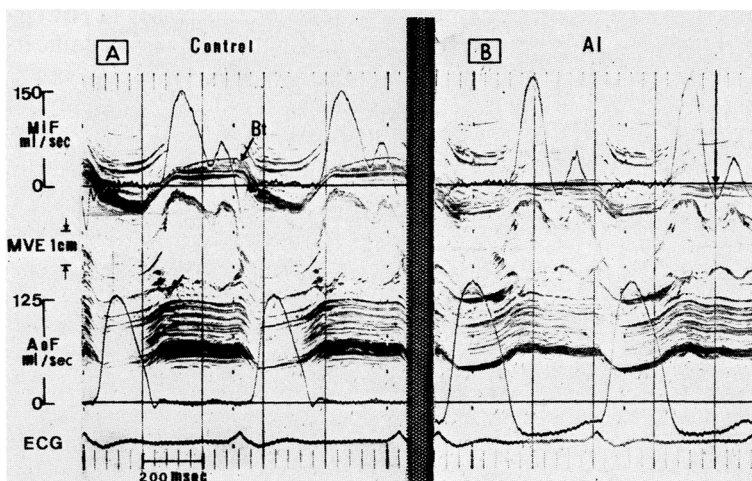


FIGURE 7. Oscillographic records of mitral flow (MiF), mitral valve echocardiogram (MVE) and aortic flow (AoF) before (A) and after (B) creation of aortic insufficiency (AI). Note the smaller opening of the mitral valve and the increased peak mitral flow after AI. Arrow (panel A) indicates the echo of the basket (Bt).

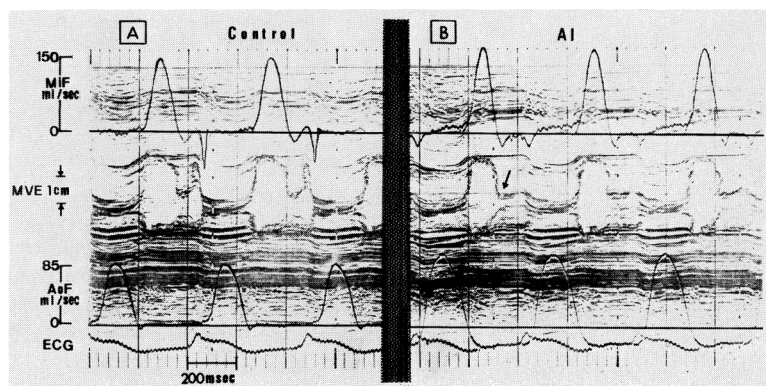


FIGURE 8. Left ventricular myocardial depression due to ischemia (note ST elevation) during control (A) resulted in redistribution of mitral flow (MiF) to early diastole similar to that during aortic insufficiency (AI) (B). Compare with figures 2–4. During acute AI, the changed atrioventricular pressure gradient (not shown) characteristic of AI, produced a premature but incomplete mid-diastolic closure of the mitral valve (arrow, panel B). MVE = mitral valve echocardiogram; AoF = aortic flow.

tract of the left ventricle. In aortic regurgitation, a low-pitched murmur was recorded during the first phase of diastole. The murmur started with the onset of rapid filling when mitral flow was accelerating to its peak, persisted almost to the end of the decelerating phase, and tended to vanish when mitral flow approached zero. The murmur was not recorded when only the aortic regurgitant jet was present.

Discussion

The hemodynamic changes we observed after the production of acute AI are consistent with expected physiologic compensations and with the results of other studies in the closed-chest dog.¹⁸ SV, LV size and LV filling pressure increased immediately. FV and aortic diastolic pressure decreased, and peak LV pressure remained the same. Because the sinoatrial node was crushed in six dogs, and because central nervous system reflexes were depressed by anesthesia, we could not observe a reflex increase in HR.¹⁸ Nevertheless, atrial pacing and the direct measurement of phasic mitral flow and other hemodynamic variables allowed us to define the mechanism by which moderately increased HR benefits patients with AI. By simultaneously recording the intracardiac phonogram and the mitral valve echocardiogram, we have provided greater insight into the genesis of the Austin Flint murmur.

Changing Pattern of Arterioventricular Flow Transport and the Significance of Increasing Heart Rate

Whereas the reduction in mitral inflow (and thus cardiac output) is expected in AI, the changes in mitral flow patterns were unexpected and interesting. Within 10 beats after the induction of AI, peak mitral flow increased and almost all transmitral flow occurred during the first half of diastole. The only late diastolic

flow was due to the atrial contraction, and it tended to be weak, possibly because the atrium was emptied early in diastole (figs. 2, 4, 8, 9 and 10). Because mitral flow is a direct function of the atrioventricular pressure difference,¹⁹ the origin of the changing flow pattern can be found in the phasic left atrial and LV pressures.

The atrial filling pressure (v-wave) increases so that at the moment of mitral valve opening, the developing atrioventricular pressure difference causes mitral flow to accelerate more rapidly and to a slightly greater level than control (figs. 2, 3, 4, 5 and 9). Thus, early in diastole the ventricle fills rapidly from both the atrium and aorta and quickly reaches the portion of its pressure-volume relation where it is relatively stiff. The combination of increasing LV pressure and decreasing left atrial pressure leads either to early-diastolic pressure equilibration or to reversal of the pressure gradient (figs. 3B and 4B). In our experience, an atrioventricular gradient reversal of less than 1 mm Hg will rapidly decelerate mitral flow. When oscillographic records are produced at low gain, such small differences are within the limits of error and resolution of the pressure traces and are not always obvious. When an early pressure crossover occurs, it may produce a small amount of mid-diastolic mitral regurgitation (figs. 2B, 3B and 7B), or late diastolic mitral regurgitation with a long PR interval.^{19, 20} Even though the aortic pressure is decreasing, the momentum of the regurgitant jet serves to maintain the retrograde flow, albeit at a slightly decreasing rate. Thus, moderate increases in HR significantly decrease the RV with only an insignificant intrusion into atrial flow transport.

Although one should use caution in applying the results of acute animal experiments to the hemodynamic behavior of the human heart in chronic AI, the

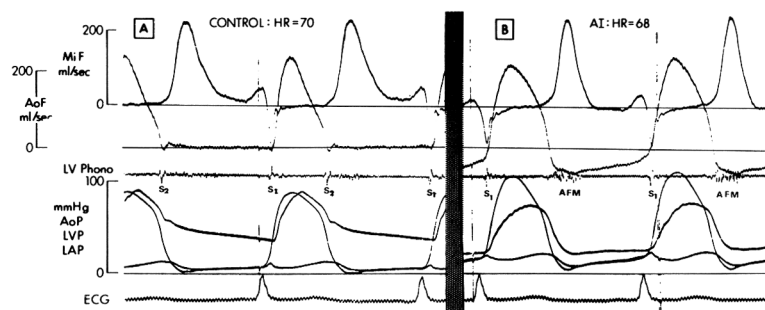


FIGURE 9. Phonocardiogram recorded in the left ventricular (LV) inflow tract, together with hemodynamic data before (A) and during (B) aortic insufficiency (AI) at the same heart rate (HR). AFM = Austin Flint murmur; S₁ = first heart sound; MiF = mitral flow; AoF = aortic flow; AoP = aortic pressure; LVP = left ventricular pressure; LAP = left atrial pressure.

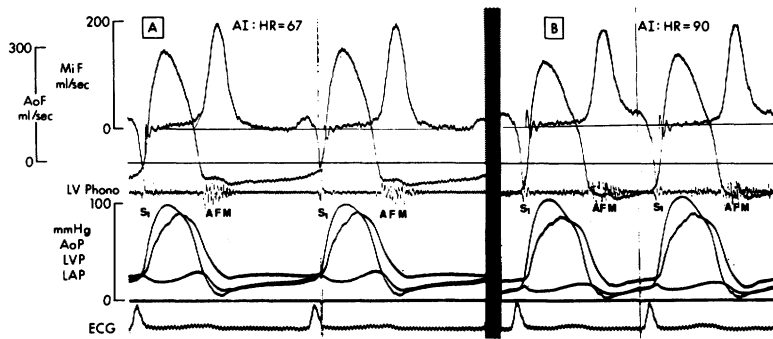


FIGURE 10. Oscillographic records during aortic insufficiency (AI) at two heart rates (HRs). The Austin Flint murmur (AFM) was registered during periods of significant antegrade mitral flow (MiF). Abbreviations as in figure 9.

observations of this study appear to be directly applicable. In the advanced stages of chronic AI, the left ventricle is stiff and the filling pressure is high.^{2,3} Thus, the mitral flow configuration recorded in these experiments may be similar to that in patients.

Clinicians have long noted that moderate tachycardia is beneficial for patients with AI, but the effect of increased HR on forward and regurgitant flows, i.e., what causes this phenomenon, have been in conflict.⁹⁻¹² Our results are in agreement with those of Judge et al.,¹² who found a significant increase in cardiac output, a decrease in LV end-diastolic pressure and no change in total regurgitant output with moderate increases in HR. Belenkie and Rademaker¹⁸ produced AI and found that the increased LV end-diastolic pressure tended to return toward control values primarily in dogs with increased HR. This is consistent with our findings (figs. 5 and 6).

Because we only paced at three rates, we cannot clearly define an optimal HR. The optimal HR will be one at which diastole is shortened without significantly affecting mitral flow. Clinically, the optimal HR will vary from patient to patient or even in the same patient as the effects of the disease change with time.

Relation of Mitral Cusp Motion to Instantaneous Diastolic Flows

The mechanism responsible for the curtailment of the full opening of mitral valve in some of our AI experiments was not completely clear. Maximal opening amplitude of the anterior cusp may have been restrained by interference from the basket located in the aortic valve, or it may have been prevented from full opening by direct impingement of the regurgitant jet. However, the increased mid-diastolic excursion of the anterior cusp toward closure (EF segment) was undoubtedly the result of the changing pattern of mitral flow after AI. We previously demonstrated that anterior cusp motion grossly followed changes in flow, and that mid-diastolic deceleration of mitral flow to zero or less was accompanied by increased closing excursion of the anterior cusp to achieve incomplete valve closure.^{14,19} Rapid deceleration of mitral flow, as can be seen in experimental animals,^{14,19} can induce premature but incomplete or nonoptimal valve closure even without AI. Such closure not preceded by a ventricular contraction may be associated with a small amount of mitral regurgitation and often accompanies a prolonged PR interval.^{19,20} In animals with experimental

AI, mitral regurgitation was seen particularly in bradycardia and with prolonged PR interval.²⁰ Premature closure of the mitral valve is also common in patients with severe acute AI.²¹

The Austin Flint Murmur

The mechanism of production of a mid-diastolic or presystolic apical low-pitched murmur in AI has been debated for years. Proposed mechanisms include incomplete opening or premature closure of the mitral valve, relative mitral stenosis, diastolic mitral regurgitation and fluttering of the cusp.²²⁻²⁸

Fortuin and Craige²⁹ used combined echo-phonocardiographic studies and concluded that the Austin Flint murmur was due to antegrade flow across the mitral valve. This was also suggested by Reddy et al.,³⁰ using intracardiac phonocardiography and high-fidelity pressure recordings. This conclusion was supported by our findings, since the intracardiac murmur, obtained from the high-fidelity micromanometer located at the LV inflow tract, was recorded only when mitral flow advanced antegradely through the valve, and subsided when mitral flow vanished or was greatly diminished (figs. 9 and 10). Using intracardiac phonocardiographic recordings, Reddy et al.³⁰ demonstrated that the typical externally recorded Austin Flint murmur was localized to the inflow of the left ventricular tract. The intracardiac murmur tended to have the characteristics of the externally recorded murmur, but had earlier onset, i.e., during the DE phase of the mitral valve echocardiogram.³⁰ In our previous experiments, the opening phase (DE) of the mitral valve echocardiogram was the time of mitral flow acceleration.^{14,19} Indeed, the LV inflow murmur started with the onset of mitral flow (figs. 9 and 10).

Fortuin and Craige²⁹ suggested that the rumbling murmur occurs during rapid closure of the mitral valve and postulated that flow velocity increases, although actual volume flow may decrease. However, Reddy and co-workers³⁰ did not find significant changes in the rate of diastolic mitral valve closure in patients with AI compared with normal subjects.

Although in the present study mitral flow velocity during the rapid filling phase increased in AI compared with control, and valve opening was moderately reduced in mid-diastole, the temporal relationship between motion and flow precluded orifice narrowing as the only mechanism of the murmur. The motion toward closure of the mitral valve, either mid-diastolic or

presystolic, always lagged behind the decelerating flow (figs. 7 and 8). When the mid-diastolic separation between the cusps was the smallest, flow velocity (measured at the level of the mitral ring) was minimal (arrow, fig. 8B), and when flow was increasing, the cusps had already moved to achieve a larger opening.

Fortuin and Craige²⁹ showed that the Austin Flint murmur had a greater intensity at a time when the two peaks of mitral flow would be predicted from the mitral valve echocardiogram, i.e., the rapid filling wave and the atrial contribution. The echocardiogram between these two peaks showed partial valve closure, and the murmur was of minimal intensity. The externally recorded murmur, which started with the opening of the mitral valve, resembled the intracardiac Austin Flint murmur recorded by Reddy et al.³⁰

Echo-phonocardiographic records from patients with AI (Criley M: personal communication) confirmed the findings of Fortuin and Craige²⁹ that the Austin Flint murmur was registered at the time of rapid mitral valve closure. Criley suggested that because the cusps formed a funnel-like conduit with the smaller diameter at the tips, mitral flow velocity was higher at the tips of the cusps compared with the ring. We measured mitral flow velocity at the proximal level of this conduit (i.e., the ring); although a higher velocity near the tips of the cusps could have played a role in the genesis of turbulence in the LV inflow tract, the pressure-flow data do not indicate a significant functional stenosis. If there were a functional stenosis (which we did not observe in the echocardiogram), the 13% increase in peak mitral volume flow after the production of AI (table 1) and the presumed reduction in area would produce a very much increased flow velocity at the cusp tips. Since the energy loss across a stenosis is proportional to the square of the velocity, we would have recorded a large increase in the atrioventricular pressure difference. This did not occur (figs. 2, 3, 4, 5 and 9), and we conclude that a significant functional stenosis did not exist in our experiments.

Shaefer et al.³¹ examined patients with Starr-Edwards mitral prostheses and suggested that the Austin Flint murmur resulted from turbulence generated by the intersection of an expanding turbulent jet flowing retrogradely through the aortic valve and a more slowly moving stream flowing antegradely across the mitral valve. In contrast, we found that the mitral flow was a rapidly rising and decelerating flow. Because their findings were based on observations in patients with prosthetic mitral valves, the relevance to the Austin Flint murmur in the natural valve can be questioned.

Since in this study there was no murmur in the absence of a regurgitant jet (fig. 9A) and there was no functional mitral stenosis in early diastole (figs. 7 and 8), we conclude that the combination of an aortic regurgitant jet and a rapidly rising and decelerating antegrade mitral flow is apparently responsible for the turbulence leading to the Austin Flint murmur. The results of this study should be applied with caution to the clinical situation. The characteristics of a murmur recorded at its origin may not be the same as those

recorded at the body surface. Furthermore, the chamber and muscle properties of a ventricle subjected to an acute volume load are different from those due to chronic stress. Thus, the timing and characteristics of the murmur recorded in this study may differ from the classic Austin Flint murmur, and may also explain why we could not record the characteristic late diastolic Austin Flint murmur.²⁹

Acknowledgment

This work could not have been done without the skilled technical help of A. Leon, P. Bon and F. Rivera. We thank M. Olivera for typing the manuscript.

References

1. Morganroth J, Perloff JK, Zeldis SM, Dunkman WB: Acute severe aortic regurgitation: pathophysiology, clinical recognition, and management. *Ann Intern Med* **87**: 223, 1977
2. Goldschlager N, Pfeifer J, Cohn K, Pepper R, Seltzer A: The natural history of acute aortic regurgitation. A clinical and hemodynamic study. *Am J Med* **54**: 577, 1973
3. Rapaport E: Natural history of aortic and mitral valve disease. *Am J Cardiol* **35**: 221, 1975
4. Henry WL, Bonow RO, Borer JS, Ware JH, Kent KM, Redwood DR, McIntosh CL, Morrow AG, Epstein SE: Observations on the optimum time for operative intervention for aortic regurgitation. I. Evaluation of the results of aortic valve replacement in symptomatic patients. *Circulation* **61**: 471, 1980
5. Henry WL, Bonow RO, Rosing DR, Epstein SE: Observations on the optimum time for operative intervention for aortic regurgitation. II. Serial echocardiographic evaluation of asymptomatic patients. *Circulation* **61**: 484, 1980
6. Clark DG, McAnulty JH, Rahimtoola SH: Valve replacement in aortic insufficiency with left ventricular dysfunction. *Circulation* **61**: 411, 1980
7. O'Rourke RA, Crawford MH: Editorial: Timing of valve replacement in patients with chronic aortic regurgitation. *Circulation* **61**: 493, 1980
8. Schlant RC: Altered cardiovascular function of rheumatic heart disease and other acquired valvular disease. In *The Heart*, edited by Hurst JW. New York, McGraw-Hill, 1978, p 970
9. Warner HR, Toronto AF: Effect of heart rate on aortic insufficiency as measured by dye-dilution technique. *Circ Res* **9**: 413, 1961
10. Rothlin M, Rutishauser W, Wirz P: Einfluss der Pulsfrequenz auf die Hamodynamik bei Aorten-kappeninsuffizienz. *Z Kreislauf-forschung* **57**: 497, 1968
11. Brawley RK, Morrow AG: Direct determinations of aortic blood flow in patients with aortic regurgitation. Effects of alterations in heart rate, increased ventricular preload or afterload, and isoproterenol. *Circulation* **35**: 32, 1967
12. Judge TP, Kennedy JW, Bennett LJ, Wills RE, Murray JA, Blackmon JR: Quantitative hemodynamic effects of heart rate in aortic regurgitation. *Circulation* **44**: 355, 1971
13. Laniado S, Yellin EL, Miller H, Frater RWM: Temporal relation of the first heart sound to closure of the mitral valve. *Circulation* **47**: 1006, 1973
14. Laniado S, Yellin EL, Kotler M, Levy L, Stadler J, Terdiman R: A study of the dynamic relations between the mitral valve echogram and phasic mitral flow. *Circulation* **51**: 104, 1975
15. Yellin EL, Kennish A, Yoran C, Laniado S, Buckley NM, Frater RWM: The influence of left ventricular filling on postextrasystolic potentiation in the dog heart. *Circ Res* **44**: 712, 1979
16. Spring DA, Rowe GG: A device for production of aortic insufficiency in intact experimental animals. *J Appl Physiol* **29**: 538, 1970
17. Yellin EL, Levy L, Bregman D, Frater RWM: Hemodynamic effects of intraaortic balloon pumping in dogs with aortic incompetence. *Trans Am Soc Artif Intern Organs* **19**: 389, 1973
18. Belenkie I, Rademaker A: Acute and chronic changes after aortic valve damage in the intact dog. *Am J Physiol* **241**: H95, 1981
19. Yellin EL, Peskin C, Yoran C, Koeningberg M, Matsumoto M, Laniado S, McQueen D, Shore D, Frater RWM: Mechanisms of

- mitral valve motion during diastole. *Am J Physiol* **241**: H389, 1981
20. Nolan SP, Fisher RD, Dixon SH, Williams WH, Morrow AG: Alterations in left atrial transport and mitral blood flow resulting from aortic regurgitation. *Am Heart J* **79**: 668, 1970
 21. Botvinick EH, Schiller NB, Wickramasekaran R, Klausner SC, Gertz E: Echocardiographic demonstration of early mitral valve closure in severe aortic insufficiency. Its clinical implications. *Circulation* **51**: 836, 1975
 22. Flint A: On cardiac murmurs. *Am J Med Sci* **44**: 29, 1862
 23. Winsberg F, Gabor GE, Hernberg JG, Weiss B: Fluttering of the mitral valve in aortic insufficiency. *Circulation* **41**: 225, 1970
 24. Herrmann GR: The Austin Flint phenomenon: an experimental and clinical study. *Am Heart J* **1**: 671, 1926
 25. Parker E, Craige E, Hood WP Jr: The Austin Flint murmur and the a-wave of the apexcardiogram in aortic regurgitation. *Circulation* **43**: 349, 1971
 26. Agnew TM, Carlisle R: Premature valve closure in patients with a mitral Starr-Edwards prosthesis and aortic incompetence. *Br Heart J* **32**: 436, 1970
 27. Lochaya S, Igarashi M, Shaffer AB: Late diastolic mitral regurgitation secondary to aortic regurgitation: its relationship to the Austin Flint murmur. *Am Heart J* **74**: 161, 1967
 28. Oliver GC Jr, Gazetopoulos N, Deucher DC: Reversed mitral diastolic gradient in aortic incompetence. *Br Heart J* **29**: 39, 1967
 29. Fortuin NJ, Craige E: On the mechanism of the Austin Flint murmur. *Circulation* **45**: 558, 1972
 30. Reddy PS, Curtiss EI, Salemi R, O'Toole JD, Griff FW, Leon DF, Shaver JA: Sound pressure correlates of the Austin Flint murmur: an intracardiac sound study. *Circulation* **53**: 210, 1976
 31. Schaefer RA, McNulty JH, Starr A, Rahimtoola SH: Diastolic murmurs in the presence of Starr-Edwards mitral prosthesis: with emphasis on the genesis of the Austin Flint murmur. *Circulation* **51**: 402, 1975

The Relationship of the First Heart Sound to Mitral Valve Closure in Dogs

LAURA F. WEXLER, M.D., GERALD M. POHOST, M.D., JOEL J. RUBENSTEIN, M.D.,
DENNIS D. O'KEEFE, M.D., MICHAEL P. VEZERIDIS, M.D., AND WILLARD M. DAGGETT, M.D.

SUMMARY The relationship between the first heart sound (S_1), mitral valve (MV) closure, and left atrial (LA) and left ventricular (LV) hemodynamics was investigated in dogs during right-heart bypass. High-speed cineradiography (~ 350 frames/sec) of lightweight clips attached to the free edge of the anterior and posterior MV leaflets permitted study of MV motion with high time resolution. LA and LV pressures were recorded, along with simultaneous phono- and echocardiograms, on a multichannel strip-chart recorder. S_1 occurred 25.5 ± 3 msec after LA-LV pressure crossover and coincided with MV closure measured either echocardiographically or cineradiographically. Atrial-ventricular (AS-VS) sequential pacing was performed to vary the interval between atrial and ventricular systole. At AS-VS intervals of 0–50 msec, S_1 and MV closure occurred simultaneously approximately 25 msec after the onset of LV systole as described above. At an AS-VS interval of 100 msec, S_1 and MV closure occurred simultaneously but closer to the onset of ventricular systole. At an AS-VS interval of 150 msec, MV closure occurred before S_1 , just preceding the onset of LV systole, and S_1 closely followed the onset of LV systole. At longer AS-VS intervals (200–300 msec), MV closure occurred at progressively earlier points in diastole without a detectable heart sound. Later in diastole, the MV reopened and then closed a second time after the onset of LV systole; S_1 was recorded with this closure. The amplitude of S_1 varied with the degree of separation of the mitral valve leaflets at the onset of LV systole. The results show that the S_1 occurs with MV closure when it follows the onset of LV systole. The timing and amplitude of S_1 are related to the degree of separation of the MV leaflets at the onset of LV systole and to the relative timing of LA and LV systole.

DESPITE 150 years of experimental study, debate about the genesis of the first heart sound (S_1) continues.¹ The two conflicting theories most frequently proposed are based on either a ventricular or a mitral valvular origin of the sound. Laennec attributed S_1 to ventricular systole, specifically vibrations developing

in the heart muscle with contraction.² Joseph Rouanet presented evidence for a valvular origin of S_1 , based on experiments with a model of the working heart and with membranes similar to valvular tissue.^{3,4}

Dock, in studies beginning in the 1930s, using both intact heart models and strips of cardiac tissue, also argued for the valvular theory.^{5,6} He found that a filled contracting ventricle tied off at the atrioventricular ring produced no sound; strips of ventricular muscle pulled taut under water produced little sound, whereas strips of chordae and cardiac valves pulled taut could be made to produce loud sounds quite easily.

Luisada and co-workers^{7–9} found that the left atrial–left ventricular pressure crossover point — which they assumed represented mitral valve closure — consistently preceded S_1 . They concluded that sudden tension development in the closed left ventricular cham-

From the Departments of Medicine, Surgery and Radiology, Harvard Medical School, and the Cardiac Unit, Surgical Cardiovascular Unit and Cardiac Radiology Unit, Massachusetts General Hospital, Boston, Massachusetts.

Supported in part by NIH grants HL-12322 and HL-12777 from the NHLBI.

Dr. Wexler's current address: Director, Coronary Care Unit, Boston City Hospital, 818 Harrison Avenue, Boston, Massachusetts 02118.

Address for correspondence: Gerald M. Pohost, M.D., Cardiac Unit, Massachusetts General Hospital, Boston, Massachusetts 02114.

Received October 20, 1980; revision accepted November 6, 1981.

Circulation **66**, No. 1, 1982.